SUMMARY · 27 OCTOBER 2026

CFD of the textured-blade turbine

What 263 simulations show about blade roughness, and what to test next

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In short. In the simulations, rough blades can increase the turbine's torque. The gain comes mostly from the moment a blade moves edge-on into the wind, not from the cup-versus-back drag that we expected to matter. With very rough blades (400 µm) the simulated torque rose by about a third, the same direction as our wind-tunnel gains. With roughness closer to our printed blades (100 µm) there was no clear change. The simulations are two-dimensional and sensitive to their settings, so they cannot confirm the mechanism on their own: a direct force measurement on a blade is the next step.

1 Why we ran these simulations

On 1 October, the FS 0.10 and FS 0.20 textured blades produced 12.8 % and 16.2 % more peak electrical power than the Plain blades (averaged over 10.2–38 m/s). We used OpenFOAM to ask two questions: how does roughness change the forces on a single blade, and how does it change the torque of the whole rotor?

2 What we simulated

All runs are two-dimensional slices through the middle of the blade, at 23 m/s unless stated, run on URI's Unity computing cluster.

Model	Cases	What was varied
Single blade	194	Angle to the wind (all the way round), wind speed (10.2, 23, 38 m/s), roughness height (smooth to 400 µm), and the printed texture shape itself
Whole rotor (3 blades)	69	Tip-speed ratio (blade-tip speed divided by wind speed), wind speed, roughness height, mesh resolution

Roughness enters the model as a uniform "sand-grain" height. Our blades' measured texture height is about 84 µm to 97 µm for FS 0.20, so the 100 µm cases are the closest to the real blades; 400 µm is a deliberately rough upper case.

3 Results

3.1 On a single blade, roughness mostly changes the edge-on forces

When the blade faces the wind cup-first or back-first, 400 µm roughness changes the drag by only about 6 %. When the blade is edge-on, the same roughness raises the drag by 45 % and the lift by 67 % (Fig. 1). At 100 µm all changes are small (10 % or less).

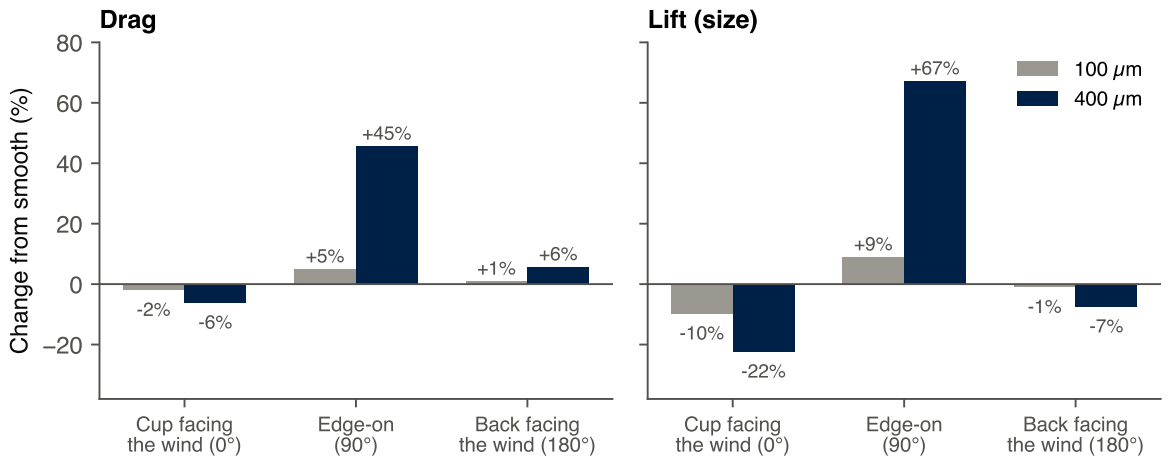


Figure 1. Change in drag and lift on a single blade from roughness, compared with a smooth blade, for three orientations (23 m/s).

3.2 On the rotor, heavy roughness raises the torque

With 400 µm roughness, the rotor's torque rose by 27 % at a tip-speed ratio of 0.10 and by 38 % at 0.12 (Fig. 2). Our turbine reached peak power near a tip-speed ratio of 0.12–0.13 in the wind tunnel. At 100 µm the changes (+8 % and −19 %) are within the simulations' noise, so we cannot say whether realistic roughness helps or hurts in this model.

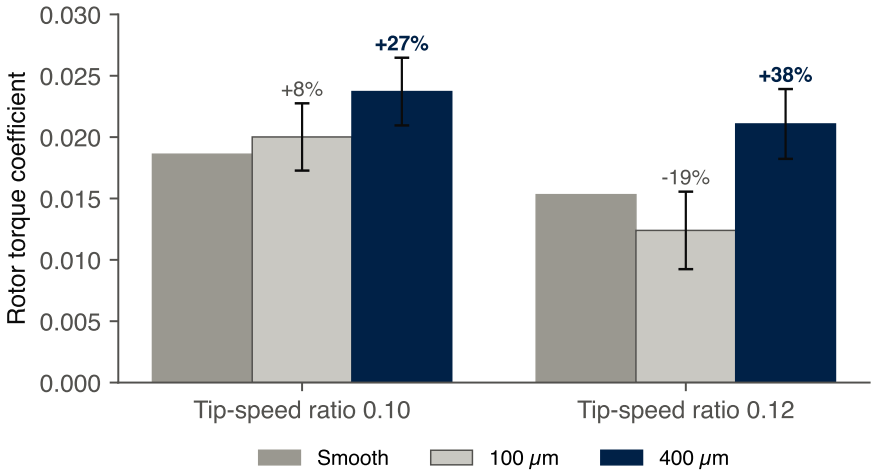


Figure 2. Rotor torque with smooth and rough blades at two tip-speed ratios (23 m/s, averaged over 45 revolutions). Error bars show the simulations' noise; a change is clear when it is larger than the bar.

3.3 The gain comes from the blade moving edge-on with the wind

Splitting the turn into positions shows where the extra torque comes from (Fig. 3). The blade moving edge-on with the wind contributes the most (+49 % of the smooth rotor's torque). Some is lost when the blade moves edge-on against the wind (−26 %). This matches the single-blade result:

the edge-on forces are the ones roughness changes most. Our earlier, simpler estimate (13 October) looked only at the cup-facing and back-facing positions, which is why it predicted a loss.

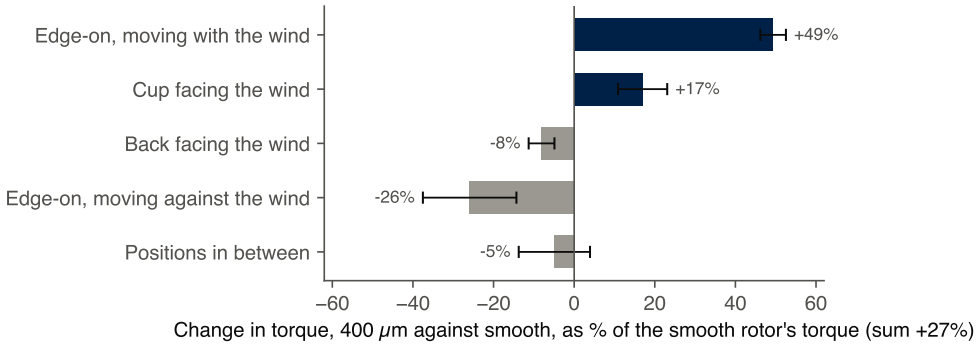


Figure 3. Where in each turn the 400 µm blades gain or lose torque, as a percentage of the smooth rotor's torque (tip-speed ratio 0.10). The parts add up to the +27 % of Fig. 2.

4 How much to trust this

- **The simulations are two-dimensional.** Flow around a curved blade is three-dimensional, and 2D simulations of this kind usually overstate drag. The results are best read as comparisons between cases rather than as absolute values.
- **The rotor result is promising but not yet firm.** The rotor's torque changes more between mesh resolutions than it does with roughness. We assumed that this affects smooth and rough runs the same way, but have not shown it.
- **The model's roughness is not our texture.** The printed fuzzy skin is not a uniform sand-grain surface, and we do not yet know which roughness height represents it.
- **One flow pattern is uncertain.** With the cup facing the wind, the simulated flow can settle into either of two stable patterns, one with nearly twice the drag of the other. Only a measurement can tell which one the real blade has.

5 Proposed next steps

1. **Measure lift and drag on single blades** (the "airfoil slice" test). An existing Plain blade and a textured blade, half in the wind, on a small two-load-cell balance under the tunnel floor, at the cup-facing, edge-on and back-facing angles. This tests the main simulation result directly. The design is ready; parts cost about $250–450.
2. **Test rougher blades.** Print FS 0.40 and FS 0.80 (the "320 µm" set) together with a fresh Plain set and repeats of FS 0.10 and FS 0.20, all with the same print settings, and test them in one session. I will ask the Print Lab for time and cost estimates.
3. **Record the rig details** the simulations had to assume: how the blades attach to the arms, which way the rotor turns, and the tunnel's test-section size.

Since the MCE labs now also use the tunnel, I am coordinating test times with Jim Byrnes and Rob D'Ambrosca.

Details, all figures and the scripts behind every number: `cfd/analysis/2026-10-15/` in the project repository.